

Wandering Black Hole Feeds on Its Own Gravitational Wake

A 35,000-solar-mass intermediate-mass black hole accretes from a dense downstream wake, revealing a new growth channel.

arXiv:2608.10719; multiwavelength evidence for off-nuclear intermediate-mass black hole accretion.

First author / year / source

[arXiv:2608.10719](#) [doi:10.48550/arXiv.2608.10719](#)



Background

Intermediate-mass black holes are thought to seed supermassive black holes, but many wander far from galactic nuclei after hierarchical mergers. Without central fuel, their growth mechanisms have remained unclear.

Direct observational evidence for gravitational-wake accretion onto off-nuclear black holes was previously elusive.

What this paper is selling

This paper reports a 35,000-solar-mass wandering black hole accreting through a gravitational wake. Multi-epoch spectroscopy reveals distinct upstream, downstream, and absorbing gas components driving rapid broad-line variability.

Key results to remember

- Broad lines, compact continuum, and variability confirm a 35,000-solar-mass black hole.
- Three gas components trace upstream flow, dense downstream wake, and optically thick absorbers.
- Changing-look broad-line variability arises from absorbers within the capture radius.

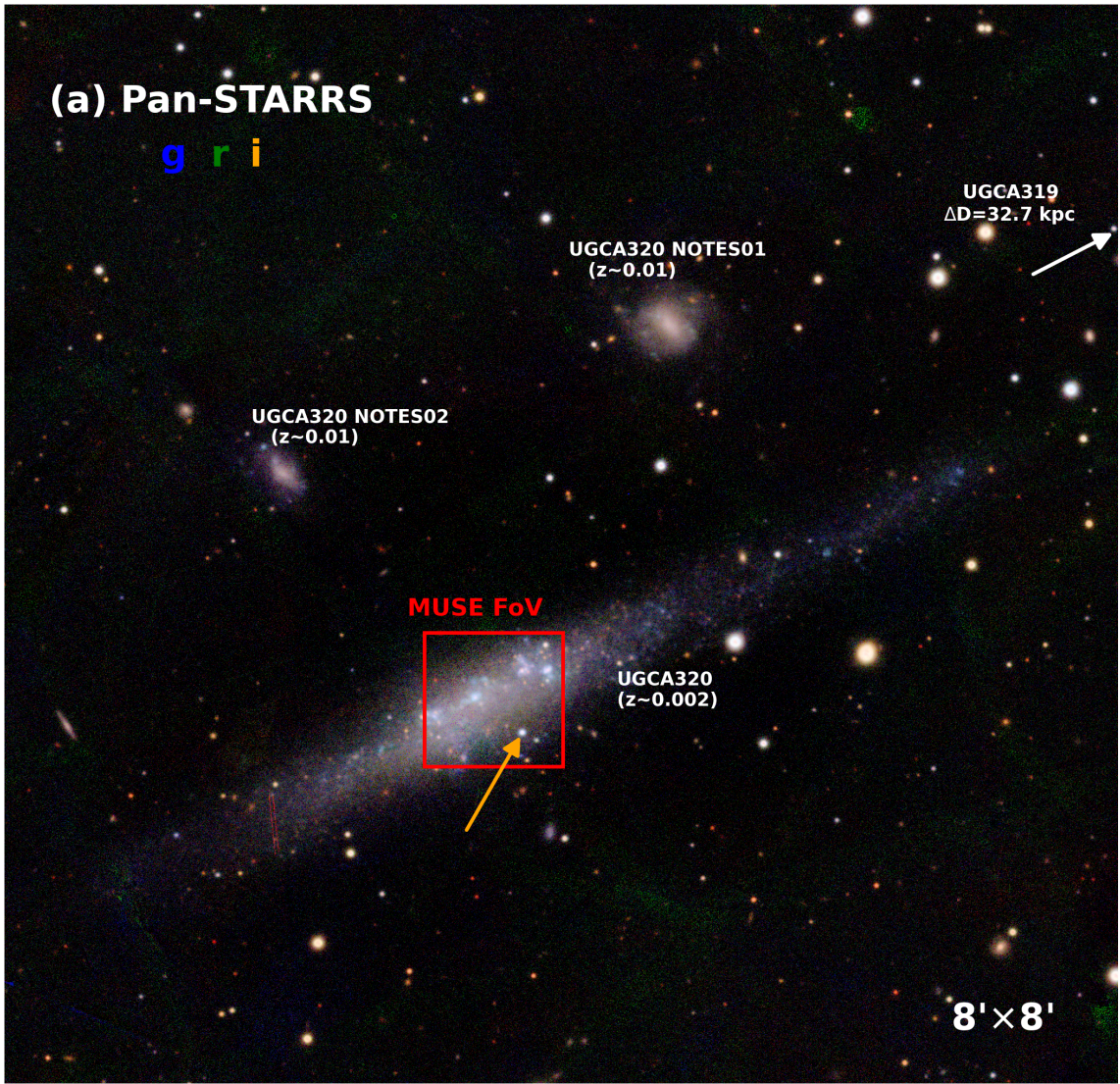


Figure 1 | Overview of UGCA 320 and UGCA320-IMBH.

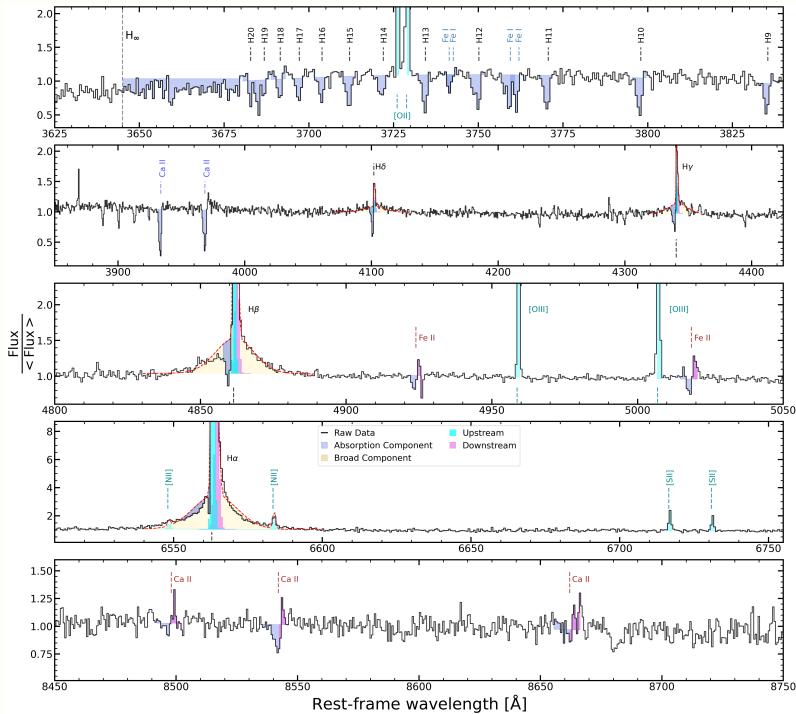


Figure 3 | X-Shooter spectrum of the UGCA320-IMBH.